In [ ]:
```
import math
# math.sqrt()

import math, random
# math.sqrt()
# random.random()
```

In [ ]:
```
from math import sqrt, factorial
# sqrt()
# factorial()
```

In [ ]:
```
from math import *
# sqrt()
# factorial()
# log()
```

In [1]:
```
from math import *

def sqrt():
    return "Hello"

print(sqrt())
```

```
Hello
```

In [3]:
```
print(sqrt(25)) #overwritten by sqrt()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[3], line 1
----> 1 print(sqrt(25))

TypeError: sqrt() takes 0 positional arguments but 1 was given
```

# random module

In [6]:
```
import random
```

## random(), uniform(a,b)

In [9]:
```
# a <= n <= b : uniform(a,b)
# 0.0 <= n < 1.0
```

In [11]:
```
random.random()
```

Out[11]:
```
0.26232597221654286
```

In [13]:
```
random.uniform(5.2,10)
```

Out[13]:
```
5.589128024073082
```

## randint(a,b), randrange([start=0],stop,[step=1])

In [16]:
```
# a <= n <= b : randint(a,b)
```

In [18]:
```
random.randint(10,15)
```

Out[18]:
```
14
```

In [20]:
```
# range(5) -> {0,1,2,3,4}
# range(1,6) -> {1,2,3,4,5}
# range(1,11,2) -> {1,3,5,7,9}
```

In [22]:
```
random.randrange(1,11,2)
```

Out[22]:
```
9
```

## choice(seq), choices(seq,*,k=1)

In [25]:
```
roll_no = list(range(101,255))
```

In [27]:
```
random.choice(roll_no)
```

Out[27]:
```
181
```

In [29]:
```
# list, tuple, string
```

In [31]:
```
random.choices(roll_no,k=5)
```

Out[31]:
```
[142, 200, 202, 247, 204]
```

In [33]:
```
t1 = tuple(range(1,15))
```

In [35]:
```
random.choices(t1, k = 3) #repeated
```

Out[35]:
```
[4, 4, 6]
```

## sample(seq,*,k=1)

In [38]:
```
random.sample(t1,k=3) #never repeats
```

Out[38]:
```
[5, 2, 1]
```

## shuffle(seq) seq : mutable sequence object only (list)

In [41]:
```
l1 = list(range(1,11))
```

In [43]:
```
l1
```

Out[43]:
```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

In [45]:
```
random.shuffle(l1)
```

In [47]:
```
print(l1)
```

```
[7, 5, 4, 2, 9, 10, 3, 1, 8, 6]
```

In [49]:
```
t1
```

Out[49]:
```
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14)
```

In [51]:
```
random.shuffle(t1)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[51], line 1
----> 1 random.shuffle(t1)

File ~\anaconda3\Lib\random.py:357, in Random.shuffle(self, x)
    354 for i in reversed(range(1, len(x))):
    355     # pick an element in x[:i+1] with which to exchange x[i]
    356     j = randbelow(i + 1)
--> 357     x[i], x[j] = x[j], x[i]

TypeError: 'tuple' object does not support item assignment
```

## seed(a = None) a : integer/float/string/None(system time)

In [54]:
```
import random
```

In [56]:
```
random.seed(10)
print(random.random())
print(random.randint(4,25))
print(random.uniform(3,7.9))
```

```
0.5714025946899135
17
5.364552205299194
```

In [58]:
```
random.seed(10)
print(random.random())
print(random.randint(4,25))
print(random.uniform(3,7.9))
```

```
0.5714025946899135
17
5.364552205299194
```

# datetime

In [61]:
```
import datetime
```

In [104…
```
# class          Purpose
# date           (year, month, day)
# time           (hour, minute, second, microsecond)
# datetime       (year, month, day, hour, minute, second, microsecond)
# timedelta      (weeks, days, hours, minutes, seconds, microseconds)
```

In [65]:
```
# p1 = Person("Darshan",101)
```

In [67]:
```
d1 = datetime.date(2026,2,24)
```

In [69]:
```
print(type(d1))
```
```
<class 'datetime.date'>
```

In [71]:
```
print(d1)
```
```
2026-02-24
```

In [73]:
```
print(datetime.date.today())
```
```
2026-03-06
```

In [75]:
```
t1 = datetime.time(7,45,23,10)
```

In [77]:
```
print(type(t1))
```
```
<class 'datetime.time'>
```

In [79]:
```
t1
```

Out[79]:
```
datetime.time(7, 45, 23, 10)
```

In [81]:
```
print(t1)
```
```
07:45:23.000010
```

In [83]:
```
dt1 = datetime.datetime(2026,2,24,7,45,23,10)
```

In [85]:
```
print(type(dt1))
```
```
<class 'datetime.datetime'>
```

In [87]:
```
print(dt1)
```
```
2026-02-24 07:45:23.000010
```

In [89]:
```
print(datetime.datetime.now())
```
```
2026-03-06 18:57:39.048316
```

In [91]:
```
import datetime
```

In [93]:
```
d1 = datetime.date(2026,2,27)
```

In [95]:
```
print(type(d1))
```
```
<class 'datetime.date'>
```

In [97]:
```
print(d1)
```
```
2026-02-27
```

In [99]:
```
nw = datetime.datetime.now()
```

In [101…
```
print(nw)
```
```
2026-03-06 18:57:43.013607
```

## strftime()

In [106…
```
s1 = datetime.datetime.strftime(nw,"%y / %b / %d - %a, %I:%M:%S:%f %p")
```

In [108…
```
print(s1)
```

```
26 / Mar / 06 - Fri, 06:57:43:013607 PM
```

## strptime()

In [113…
```
x = datetime.datetime.strptime(s1,"%y / %b / %d - %a, %I:%M:%S:%f %p")
```

In [115…
```
print(type(x))
```

```
<class 'datetime.datetime'>
```

In [117…
```
print(x)
```

```
2026-03-06 18:57:43.013607
```

In [119…
```
d1 = datetime.date(2025,1,1)
d2 = datetime.date(2022,1,1)
```

In [121…
```
d2 - d1
```

Out[121…
```
datetime.timedelta(days=-1096)
```

In [123…
```
nw
```

Out[123…
```
datetime.datetime(2026, 3, 6, 18, 57, 43, 13607)
```

In [125…
```
td = datetime.date.today()
```

In [127…
```
td
```

Out[127…
```
datetime.date(2026, 3, 6)
```

In [129…
```
td + datetime.timedelta(weeks=2)
```

Out[129…
```
datetime.date(2026, 3, 20)
```

In [131…
```
t1 = datetime.time(7,30,23,78)
```

In [133…
```
t1 + datetime.timedelta(hours = 1)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[133], line 1
----> 1 t1 + datetime.timedelta(hours = 1)

TypeError: unsupported operand type(s) for +: 'datetime.time' and 'datetime.timed
elta'
```

# math module

In [136…
```
import math
```

In [138…
```
math.pi
```
Out[138…
```
3.141592653589793
```

In [140…
```
math.e
```
Out[140…
```
2.718281828459045
```

In [142…
```
math.sqrt(25)
```
Out[142…
```
5.0
```

In [144…
```
math.pow(2,3)
```
Out[144…
```
8.0
```

In [146…
```
2**3
```
Out[146…
```
8
```

In [148…
```
math.ceil(4.2)
```
Out[148…
```
5
```

In [150…
```
math.floor(4.7)
```
Out[150…
```
4
```

In [152…
```
round(4.2)
```
Out[152…
```
4
```

In [154…
```
round(4.7)
```
Out[154…
```
5
```

In [156…
```
abs(-10)
```
Out[156…
```
10
```

In [158…
```
math.fabs(-10)
```
Out[158…
```
10.0
```

In [160…
```
math.gcd(12,9)
```
Out[160…
```
3
```

In [162…
```
math.lcm(12,9)
```
Out[162…
```
36
```

In [164…
```
math.sin(math.pi/2)
```
Out[164…
```
1.0
```

In [166…
```
math.cos(math.pi/4)
```
Out[166…
```
0.7071067811865476
```

In [168…
```
math.tan(math.pi/4)
```
Out[168…
```
0.9999999999999999
```

In [170…
```
math.sin(math.radians(90))
```
Out[170…
```
1.0
```

In [172…
```
math.log(8,2)
```
Out[172…
```
3.0
```

In [174…
```
math.log10(100)
```
Out[174…
```
2.0
```

In [176…
```
math.log(10)
```
Out[176…
```
2.302585092994046
```

In [178…
```
math.exp(1)
```
Out[178…
```
2.718281828459045
```

In [180…
```
math.hypot(3,4)
```
Out[180…
```
5.0
```

In [182…
```
math.inf
```
Out[182…
```
inf
```

In [184…
```
math.isinf(math.inf)
```
Out[184…
```
True
```

In [186…
```
%%writefile file1.py
def greet():
    print("Hello World.!!")

def greet_name(name):
    print("Hello",name)
```
```
Overwriting file1.py
```

In [188…
```
import file1
```

In [190…
```
file1.greet()
```
```
Hello World.!!
```

In [192…
```
file1.greet_name("Students")
```

```
Hello Students
```

In [194…
```python
import file2
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
Cell In[194], line 1
----> 1 import file2

ModuleNotFoundError: No module named 'file2'
```

In [196…
```python
import sys
```

In [198…
```python
for i in sys.path:
    print(i)
```

```
D:\AY-2526\1. Python Programming\Python_Programs\jupyter\Module_4A
C:\Users\mania\anaconda3\python312.zip
C:\Users\mania\anaconda3\DLLs
C:\Users\mania\anaconda3\Lib
C:\Users\mania\anaconda3

C:\Users\mania\anaconda3\Lib\site-packages
C:\Users\mania\anaconda3\Lib\site-packages\win32
C:\Users\mania\anaconda3\Lib\site-packages\win32\lib
C:\Users\mania\anaconda3\Lib\site-packages\Pythonwin
C:\Users\mania\anaconda3\Lib\site-packages\setuptools\_vendor
```

In [200…
```python
sys.path.append("C:\\Users\\mania\\OneDrive\\Desktop")
```

In [202…
```python
import file2
```

In [204…
```python
file2.add(1,5)
```

Out[204…
```
6
```

In [206…
```python
import random
student_roll_list = list(range(101,255))

def random_select(n):
    return random.sample(student_roll_list,k=n)
```

In [208…
```python
random_select(5)
```

Out[208…
```
[104, 153, 219, 226, 172]
```